

Chemistry Ch 4.8: Noble Gases (Group 18)

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Teen khasoosiyat

1. **Inert**: ye kisi se amal nahin karte.
2. **Mono-atomic**: ye **akele atom** ki shakl mein rehte hain. Baqi gases jori (O_2 , N_2) banati hain, ye nahin.
3. **Bhari hui bairooni shell**: helium ke paas **2**, baqi sab ke paas **8**.

Gas	Bairooni configuration
Helium (He)	$1s^2$ (sirf 2 , kyunke pehli shell mein 2 hi aate hain)
Neon (Ne)	$2s^2 2p^6$
Argon (Ar)	$3s^2 3p^6$

Inert hone ki wajah, aur do nateeje

Jab bairooni shell **poori bhari** ho to atom ko na electron **denay** ki zaroorat hai na **lenay** ki. Wo pehle hi mustahkam hai.
Isi se dono baqi baatein nikal aati hain: wo **ion nahin** banate, aur wo doosre atom se **jorta** bhi nahin, is liye **akela** rehta hai, yani mono-atomic.

Isi be-amli ka faida

Helium ghubbaron mein, kyunke wo halki hai aur **aag nahin pakarti**. Neon roshni wale signon mein.
Aur argon bulb ke andar, taake filament jal na jaye.
Yani in ka faida hi ye hai ke ye **kuch nahin karte**.

Quiz

1. Give the three characteristic properties of the noble gases, and write the outer electronic configuration of helium, neon and argon.
2. Explain why the noble gases are both **inert** and **mono-atomic**. Why does helium have 2 outer electrons rather than 8?